bucket slows because of the tightening rope, the water continues to spin—spinning with respect to absolute space—and that's why its surface continues to be concave. And so, whereas relative motion between the water and the bucket cannot account for the observations, relative motion between the water and absolute space can. Space itself provides the true frame of reference for defining motion.

The bucket is but an example; the reasoning is of course far more general. According to Newton's perspective, when you round the bend in a car, you feel the change in your velocity because you are accelerating with respect to absolute space. When the plane you are on is gearing up for takeoff, you feel pressed back in your seat because you are accelerating with respect to absolute space. When you spin around on ice skates, you feel your arms being flung outward because you are accelerating with respect to absolute space. By contrast, if someone were able to spin the entire ice arena while you stood still (assuming the idealized situation of frictionless skates)—giving rise to the same relative motion between you and the ice—you would not feel your arms flung outward, because you would not be accelerating with respect to absolute space. And, just to make sure you don't get sidetracked by the irrelevant details of examples that use the human body, when Newton's two rocks tied together by a rope twirl around in empty space, the rope pulls taut because the rocks are accelerating with respect to absolute space. Absolute space has the final word on what it means to move.

But what is absolute space, really? In dealing with this question, Newton responded with a bit of fancy footwork and the force of fiat. He first wrote in the *Principia* "I do not define time, space, place, and motion, as [they] are well known to all,"[3] sidestepping any attempt to describe these concepts with rigor or precision. His next words have become famous: "Absolute space, in its own nature, without reference to anything external, remains always similar and unmovable." That is, absolute space just is, and is forever. Period. But there are glimmers that Newton was not completely comfortable with simply declaring the existence and importance of something that you can't directly see, measure, or affect. He wrote,

It is indeed a matter of great difficulty to discover and effectually to distinguish the true motions of particular bodies from the apparent,